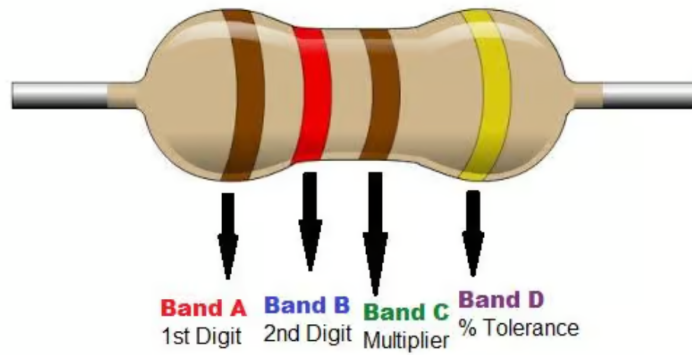


Identification of Electronic Components

Resistor



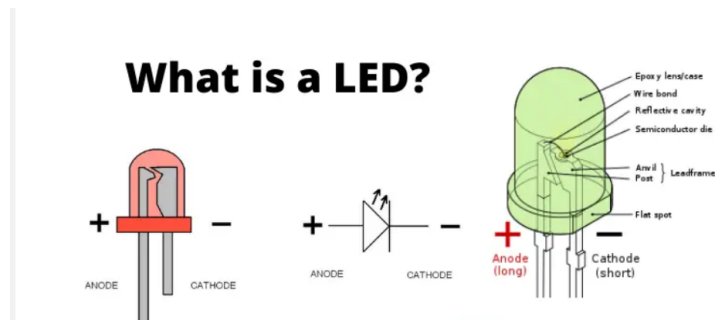
Resistor: Commonly used in electronics and IoT projects.

Capacitor



Capacitor: Commonly used in electronics and IoT projects.

LED



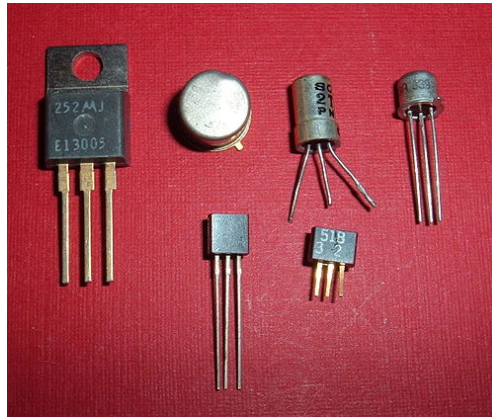
LED: Commonly used in electronics and IoT projects.

Diode



Diode: Commonly used in electronics and IoT projects.

Transistor



Transistor: Commonly used in electronics and IoT projects.

Push Button



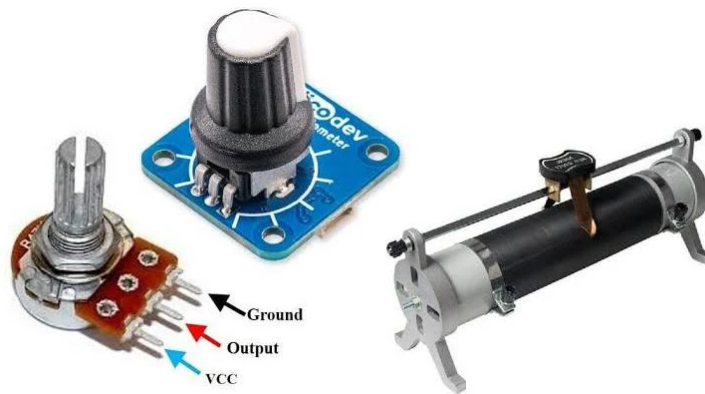
Push Button: Commonly used in electronics and IoT projects.

Buzzer



Buzzer: Commonly used in electronics and IoT projects.

Potentiometer



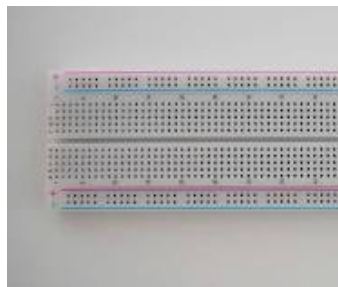
Potentiometer: Commonly used in electronics and IoT projects.

Arduino Uno



Arduino Uno: Commonly used in electronics and IoT projects.

Breadboard



Breadboard: Commonly used in electronics and IoT projects.

MAJOR COMPONENTS USED IN ELECTRONICS

1. Resistor

- *A resistor is used to limit the flow of electric current in a circuit.
- *It protects electronic components from excessive current.
- *Resistance value is identified using color bands.

2. Capacitor

- *A capacitor stores electrical energy temporarily.
- *It is used for filtering and smoothing voltage fluctuations.
- *Capacitors are available in various capacitance values.

3. LED (Light Emitting Diode)

- *An LED emits light when current flows through it.
- *It consumes less power than conventional bulbs.
- *LEDs are widely used as indicators and displays.

4. Diode

- *A diode allows current to flow in only one direction.
- *It is commonly used for rectification and protection circuits.
- *The cathode side is marked with a stripe.

5. Transistor

- *A transistor acts as an electronic switch or amplifier.
- *It controls a large current using a small input current.
- *It is a key component in modern electronic devices.

6. Push Button

- *A push button is a simple input device used to control circuits.
- *It completes the circuit when pressed.
- *Push buttons are commonly used in user interfaces.

7. Buzzer

- *A buzzer converts electrical energy into sound.
- *It is used for alarms and notification systems.
- *Buzzers can be active or passive types.

8. Potentiometer

- *A potentiometer is a variable resistor.
- *It is used to adjust voltage and current levels.
- *Common applications include volume and brightness control.

9. Arduino Uno

- *Arduino Uno is a microcontroller development board.
- *It is used for programming and controlling electronic projects.
- *It supports various sensors and actuators.

10. Breadboard

- *A breadboard is used to build circuits without soldering.
- *It allows easy testing and modification of connections.
- *Breadboards are commonly used for prototyping projects.